Loss Function Design for Wrapped Phase Fitting in InSAR Deep Learning Network Training

Chunjing Chen, Chisheng Wang, and Haidong Yu

***Abstract*— In recent years, the application of deep learning in the field of interferometric synthetic aperture radar (InSAR) measurement has gradually increased. Unlike the real-value data typically handled by conventional deep learning methodologies, InSAR technology grapples with complex-value data. Specifically, the phase values of this complex data are constrained within the interval $(-\pi, +\pi]$. However, in deep learning frameworks, the calculation of loss functions often neglects the mathematical constraint that phase differences should also adhere to the $[-\pi, \pi]$ range. This oversight impedes the model's ability to accurately fit the target data, thereby diminishing training efficiency. To address this challenge, we introduce three foundational loss functions tailored for fitting wrapped phase values: cosine similarity-based (La), mean square error (Lb) for wrapped phase, and ensemble coherence-based (Lc). By combining these approaches, we formulate a total of seven distinct loss functions. To identify the most effective one, we evaluated their performance using a temporal convolutional network (TCN) as a test scenario. Our findings reveal that the optimal combination function, Labc, achieves a 72.78% accuracy rate. Compared to other function with over 72% accuracy, Labc reduces the required iterations for model convergence by at least 30% and effectively delineates terrain feature boundaries.**

***Index Terms*— Deep learning, interferometric synthetic aperture radar (InSAR), loss function, wrapped phase.**

I. Introduction

INTERFEROMETRIC synthetic aperture radar (InSAR) is a powerful technique widely employed across various domains [1], [2], including surface deformation monitoring, crustal movement analysis, and disaster management. It captures surface deformation information by monitoring phase variations in radar signals over both temporal and spatial dimensions. There have been many advanced InSAR algorithms developed in this field over the past decades, enabling increasingly precise and detailed deformation measurements. In recent years, the utilization of deep learning techniques in earth science has witnessed a steady rise, offering innovative solutions to challenges encountered in InSAR applications, particularly in interferometric phase filtering or optimization. Convolutional neural networks (CNNs) and recurrent neural networks (RNNs) have emerged as key methodologies in tackling these challenges [3], [4], [5], [6]. For instance, researchers have proposed CNN-based models to monitor coseismic deformation in interferograms [7], [8] and deep learning approaches for change point detection in InSAR time series [9]. Additionally, methods combining CNN and RNN have been introduced to effectively filter noise from interferometric phases, thereby enhancing estimation accuracy and reliability [6]. These advancements underscore the significant potential of deep learning in InSAR applications, presenting valuable methodologies for progression of this field.

In the studies mentioned above, the choice of loss function significantly influences the effectiveness of model processing and interpretation on interferometric phase data. Therefore, selecting an appropriate loss function is crucial when training a deep learning framework. During each iteration of training, the model updates its parameters using gradient information derived from the loss function. This iterative process aims to minimize the discrepancy between the predicted and actual values, facilitating learning. However, in scenarios involving wrapped phase values calculation, standard loss functions designed for real data often overlook the unique mathematical characteristics of phase values, resulting in inaccurate loss value computations.

In the process of calculating the loss value, it is often challenging to ensure that the resulting phase difference value remains within the range of $(-\pi, +\pi]$. Once this range is exceeded, the outcome may diverge significantly from the actual value, thereby impacting subsequent phase calculations. Addressing these challenges, this letter introduces three fundamental loss functions and explores their combination, resulting in a total of seven distinct loss functions, which are suitable for handling regression-like problems involving wrapped phase values. These functions accurately compute the difference between two phase values. The three foundational functions include cosine similarity-based (La), mean square error (Lb), and ensemble coherence-based (Lc).

Specifically, Lb incorporates a wrapping criterion in its difference computation process, ensuring adherence to the range of $(-\pi, +\pi]$. Meanwhile, Lc utilizes ensemble coherence (EC), which is employed to assess the residual phase following model fitting in distributed scatterer interferometry

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TABLE I
INFORMATION OF SEVEN LOSS FUNCTIONS

Functions	Message
La	Based on cosine similarity
Lb	A special MSE for wrapped phase values
Lc	Based on ensemble coherence
Lab	Combination of La and Lb
Lac	Combination of La and Lc
Lbc	Combination of Lb and Lc
Labc	Combination of La, Lb and Lc

(DSI), a crucial metric for the evaluation of interferometric quality [1], [10]. Based on these three foundational functions, we systematically combined them to form seven distinct functions used in our experiments: La, Lb, Lc, Lab, Lac, Lbc, and Labc, as detailed in Table I.

In addition, a deep learning framework based on temporal convolutional network (TCN) [11] was introduced to emulate the phase linking [12], [13] process of wrapped phase, serving as the test scenario in this study. We conducted seven independent experiments and established criteria for similarity judgment as evaluation metrics to ensure the reliability and validity of our research results.

II. METHODOLOGY

This chapter introduces the design of loss functions, the construction of the testing scenario, and criteria for evaluating these functions.

A. Loss Functions

Interferograms typically appear in complex format, where phase values are wrapped within the range of $[-\pi, \pi]$. Conventional loss functions calculate the differences of phase values numerically but often overlook the requirement that these differences should also adhere to the range of $[-\pi, \pi]$. Consequently, we have developed customized loss functions specifically tailored for training neural networks on wrapped phase values.

1) Loss Function of Cosine Similarity (La): La rearranges the two arrays into a 1-D vector and computes the cosine similarity between them. The loss value based on cosine similarity is calculated using the following formula:

$$\mathrm{La} = 1 - \frac{1}{N} \sum \frac{\Phi_1 \cdot \Phi_2}{\max(\| \Phi_1 \|_2, \epsilon) \cdot \max(\| \Phi_2 \|_2, \epsilon)} \tag{1}$$

where Φ_1 and Φ_2 denote the two vectors involved in the computation, and $\| \cdot \|_2$ denotes the 2-norm, representing the length of the vector. ϵ is used to prevent division by zero in the denominator and defaults to 10^{-8}. N represents the number of elements contained in the sample.

Cosine similarity ranges from -1 to 1. Equation (1) computes the mean cosine similarity within a batch and then subtracts this mean from 1 to obtain the loss value. This method prevents negative loss values, and a value closer to 0 indicates better model performance.

2) MSE for Wrapped Phase (Lb): Mean square error (MSE) has the capability to magnify errors and offer a more robust gradient signal for model optimization. Taking into account the benefits of MSE and the particular characteristics of the experimental data, we incorporate a correction step in the residual calculation of the wrapped phase values

$$\mathrm{Lb} = \frac{1}{N} \sum \Delta\Phi^2 \tag{2}$$

$$\Delta\Phi = (\Phi_2 - \Phi_1 + \pi)\%(2\pi) - \pi. \tag{3}$$

Equation (2) delineates the methodology for computing the MSE of the residual difference $\Delta\Phi$ in the wrapped phase. Here, N denotes the number of elements in the sample, where Φ_1 and Φ_2 represent the target data and the result of model, respectively. $\Delta\Phi$ is depicted in (3). The operator symbol signifies the remainder when two numbers are divided. In both formulas, the range of $\Delta\Phi$ is $(-\pi, \pi]$.

3) Loss Function of Ensemble Coherence (Lc): EC acts as a metric for quantifying the phase residual between estimated and true values, serving as an indicator to evaluate the overall quality of DSI. The loss function derived from EC is expressed in the following equation:

$$\mathrm{Lc} = 1 - \frac{1}{N} \| \sum \left(e^{j\Phi_1} \cdot e^{-j\Phi_2} \right) \|_2 \tag{4}$$

where $e^{j\Phi_1}$ denotes the complex array corresponding to the target data, $e^{-j\Phi_2}$ denotes the complex conjugate array of the result from the model, and the complex array containing the residual phase is obtained through the dot multiplication of the two. $\| \cdot \|_2$ denotes the two-norm and N is the number of elements in the sample. The value of loss_ec falls within the range [0, 1], where closer to 0 indicates that the result of the model is closer to the target data.

4) Composite Functions of Three Basic Loss Function: In (1)–(4) presented earlier, it is evident that La, Lb, and Lc focus, respectively, on computing the similarity in vector direction of phase values, numerical differences, and residual phase discrepancies. We combine these basic functions to derive composite loss functions, whose formulations are detailed as follows:

$$\mathrm{Lab} = \mathrm{La} + \mathrm{Lb} \tag{5}$$

$$\mathrm{Lac} = \mathrm{La} + \mathrm{Lc} \tag{6}$$

$$\mathrm{Lbc} = \mathrm{Lb} + \mathrm{L} \tag{7}$$

$$\mathrm{Labc} = \mathrm{La} + \mathrm{Lb} + \mathrm{Lc}. \tag{8}$$

To evaluate the performance of the aforementioned functions, a deep learning framework simulating phase linking is employed as the experimental test scenario.

B. Test Scenario

The phase linking method enhances interferometric phase estimation for distributed scatterers in InSAR observations, improving measurement accuracy. We used a TCN structure to simulate phase linking [14], utilizing temporal and spatial data of interferograms to estimate wrapped phase values and reduce noise. Fig. 1 illustrates the TCN structure. In this study, TCN serves as the test scenario for evaluating the proposed

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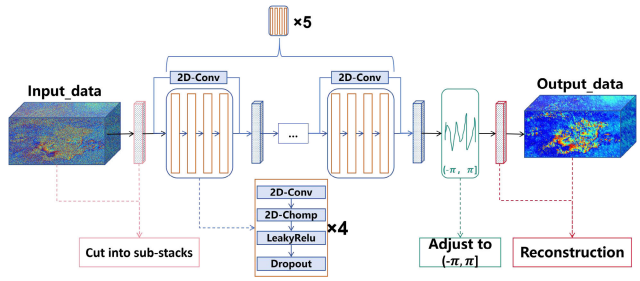

Fig. 1. TCN structure used in the experiments.

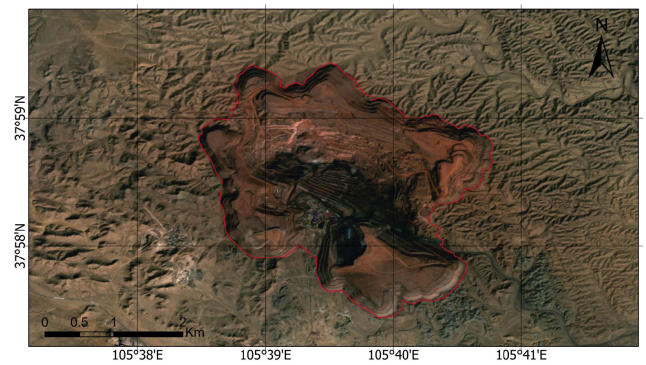

Fig. 2. Schematic of the experimental area. The base map is sourced from Map World, with the main experimental area delineated by the red polygon.

loss functions, with parameters optimized based on loss values to achieve rapid and accurate wrapped phase values estimation.

Considering the mathematical characteristics of the wrapped phase, it is imperative to assess and adjust the results before the model outputs them. This adjustment process, as illustrated in (9), involves evaluating the output denoted by $\phi(\text{out})$

$$\phi(\text{out}) = \begin{pmatrix} \phi(\text{out}), & -\pi < \phi(\text{out}) \leq \pi \\ \phi(\text{out}) - 2\pi, & \phi(\text{out}) > \pi \\ \phi(\text{out}) + 2\pi, & \phi(\text{out}) \leq \text{Ü} - \pi \end{pmatrix}. \quad (9)$$

In forthcoming experiments, we will employ seven loss functions to train the TCN model and compare the training effects of each model.

C. Evaluation Criteria

To evaluate the proposed loss functions, we calculate the mean absolute error between each model's output and the target data. Smaller differences indicate better performance. We also consider convergence iterations as a metric for comparing loss function effectiveness; fewer iterations imply faster convergence.

Additionally, overall accuracy was employed to gauge the similarity between the output of models and the target data. This accuracy was determined by computing the absolute difference between corresponding pixels. Pixels with differences below a threshold ($(2\pi/12)$) were deemed similar. The threshold is determined based on mapping phase values onto a color bar divided into 12 segments, where each segment represents similar colors within the range of $[-\pi, \pi]$. Ultimately, the overall accuracy is defined as the proportion of similar pixels to the total number of pixels.

III. Experience

Seven independent experiments were conducted under identical test scenarios. This section details the experimental area, datasets, and the procedures involved.

A. Datasets

The experimental area is an open-pit mining region in southwest Inner Mongolia, China, covering approximately 1.69 km^2. We introduce the experimental area in Fig. 2.

Following the identification of the experimental area, we collected 31 Sentinel-1A SLC images that cover from June 5, 2021, to November 15, 2022, with a temporal interval of 12 days, detailed in Table II. After processing, 30 interferograms were obtained, each sized 500 × 1500 pixels.

B. Experimental Process

In the preprocessing stage, the image with the middle date was selected as the main image. External data, such as DEM and precise orbits, were employed for coregistration to generate interferograms.

Prior to commencing the experiments, phase values were computed for each pixel in both the raw interferograms and the phase linking-processed interferograms. These phase values were converted from complex numbers to floating-point numbers within the range of $(-\pi, \pi]$. Subsequently, the computed interferograms were stacked along the temporal dimension to create the training datasets "Input_data" and "Target_data."

The datasets were segmented into smaller subsequences and padded to ensure overlap between adjacent segments. This process yielded 300 pairs of subsequences, each sized 64 × 64 × 30. These pairs were utilized for training the TCN model to facilitate learning of the phase linking process.

We prioritized sufficient training iterations to ensure model convergence. During testing, we used different loss functions to quantify the disparity between the model result and the target data, thereby facilitating optimization via back-propagation. Seven TCN models, each trained for 800 epochs, were configured with identical parameters but different loss functions. The input data, initially in complex form, were transformed into floating-point phase values, resulting in a single input channel. The learning rate was set to 0.01, and the Adam optimizer was employed. The input data, output data from seven models, and target data all consist of 3-D arrays containing phase values from 30 interferograms. Their shapes, sizes, and corresponding positions remain consistent. The complete experimental procedure is depicted in Fig. 3.

After training, we plotted the learning curves of the model and compared the results using the evaluation criteria from Section II-C. This allowed us to assess the performance of each loss function.

IV. Results and Discussion

In the same testing scenario and under identical parameter conditions, we compared and evaluated the loss functions based on convergence speed, accuracy of output data, and mean absolute error.

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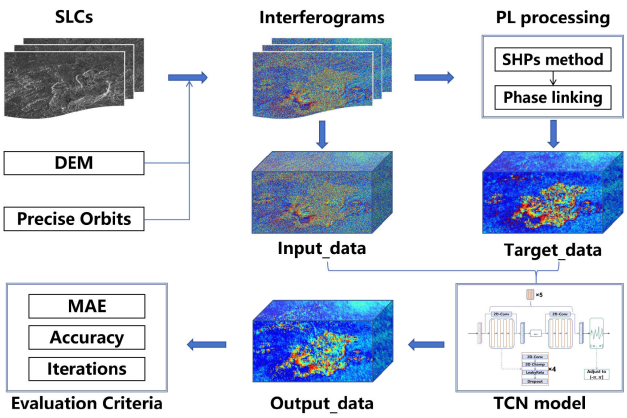

Fig. 3. Flowchart of experiment.

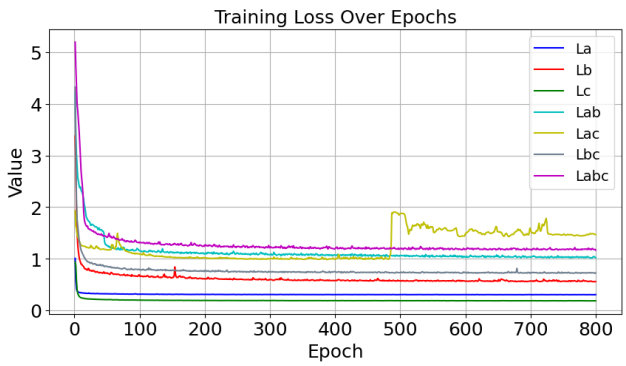

Fig. 4. Learning curves of loss functions.

TABLE II
DETAILED INFORMATION OF THE SLCS

Label	Message
Sensor	Sentinel-1A
Temporal Coverage	5 June 2021 - 15 November 2022
Image Number	31
Direction	Ascending
Beam Mode	IW
Polarization	VV+VH

A. Learning Curves

We assessed convergence through changes in the learning curves during training, plotting loss values over iterations. As depicted in Fig. 4, during the initial 100 training epochs, most functions facilitate rapid model convergence, followed by a gradual stage. Notably, the curves for La and Lc flatten within the first 100 epochs, while Lb, Lab, and Lac exhibit occasional significant fluctuations.

Subsequently, we selected models corresponding to the flattened segments of the learning curves and quantitatively compared their output results using the established criteria. As shown in Table III, we observe that models using La and Lc loss functions achieve convergence within 100 training epochs, while models employing Lac and Labc converge after 350 epochs. Other loss functions require 500 epochs for convergence.

TABLE III
QUANTITATIVE RESULTS

Function	MAE	Accuracy (%)	Converged epochs
La	1.58	19.82	100
Lb	0.80	71.72	500
Lc	2.40	2.67	100
Lab	0.96	66.34	500
Lac	66.42	1.30	350
Lbc	0.80	72.24	500
Labc	0.76	72.78	350

B. Quantitative Results of Models

However, solely judging convergence speed by learning curve changes may be insufficient. The numerical difference between the current model output and target data, influenced by the calculation method, affects feature extraction and learning speed. Thus, we calculated the accuracy and the mean absolute errors of models, which are essential for accurately estimating wrapped phase values.

We computed the absolute errors between the output results of each model and the corresponding target data on a per-pixel basis. A threshold was set to determine whether the current pixel is similar to the target data pixel. Finally, we aggregated the mean absolute error for each model and calculated the average accuracy, which representing the proportion of similar pixels to the total number of pixels. These metrics are presented in Table III as "MAE" and "accuracy."

As shown in Table III, while La, Lc, and Lac converge quickly, their results significantly deviate from the target data, evident from the large errors (1.58, 2.4, and 66.42) and low accuracy rates (19.8%, 2.67%, and 1.30%). In contrast, Lb, Lbc, and Labc demonstrate superior performance with similar numerical outcomes. Notably, Labc achieves the fastest convergence, the lowest mean absolute error, and the highest accuracy. This superior performance can be attributed to Labc combining the rapid convergence of La, the error sensitivity of Lb, and the phase difference calculation of Lc.

C. Boundary Comparison of Output Results

In the previous description, we identified that the Lbc and Labc loss functions achieved the best performance, exhibiting similar accuracy. Therefore, we performed a color mapping of the model results, which are floating-point phase values within the threshold range of $(-\pi, +\pi]$. These were compared against the raw interferograms (input data) and the phase linking processed interferograms (target data).

Results for the time step of 22 were selected for illustration. Fig. 5 highlights a portion of the main surface features, with Fig. 5(a)–(d) representing the input data, output data of Lbc, output data of Labc, and target data, respectively. It is evident that the results of both Lbc and Labc closely resemble the target data, effectively filtering the input data.

Additionally, Fig. 6 presents the results over smaller regions, allowing for a detailed examination of boundary features. In Fig. 6(a) and (b), the output results of Lbc and Labc are shown, respectively. Although the results are very similar, the color transitions in Fig. 6(b) are more natural, demonstrating greater sensitivity to phase values near zero and preserving

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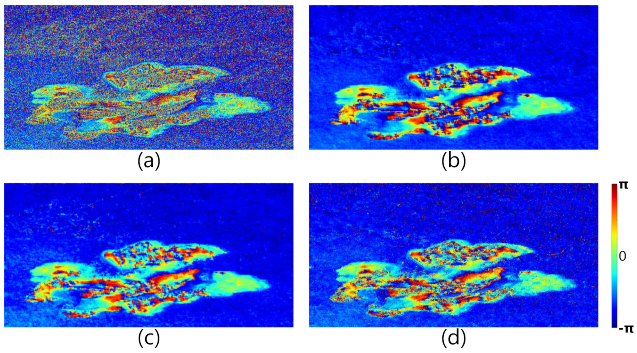

Fig. 5. Comparison of (a) input data, (b) output result of Lbc, (c) output result of Labc, and (d) target data.

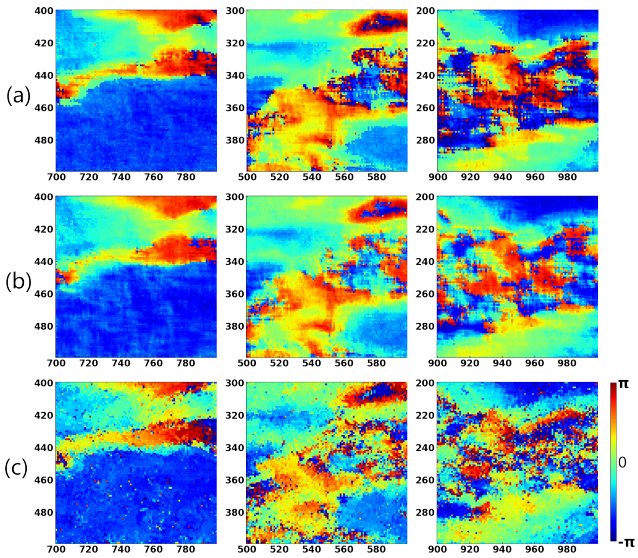

Fig. 6. Boundary comparison of output results and target data. (a) Output result of Lbc. (b) Output result of Labc. (c) Target data.

more boundary details. Conversely, the boundaries in Fig. 6(a) are sharper but are more likely to overlook phase values near zero. When compared with the target data in Fig. 6(c), it is evident that Fig. 6(b) preserves more details, indicating that Labc helps the model achieve boundary delineations that are closer to the target data.

Overall, Labc demonstrates superior performance in terms of model convergence rate, mean absolute error, and accuracy. It excels at delineating boundaries while preserving more detailed information, making it the best-performing loss function among the seven evaluated.

V. Conclusion

This letter introduces three basic loss functions, which are then combined to form seven different loss functions, to enhance the robustness of wrapped phase values fitting in deep learning training. A TCN-based network is employed as the experimental scenario, and seven independent experiments with identical parameter settings were conducted to evaluate the performance of these loss functions.

The results indicate that the most effective loss functions are Lbc and Labc, with the latter preserving more boundary detail information. Therefore, the Labc loss function, which combines cosine similarity (La), mean squared error for wrapped phase (Lb), and ensemble coherence (Lc), proves to be the best performer in the experiments.

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